



HTML Fundamentals

The Building Blocks of Every Website

Cardiff University · Web Development

Module 1 · 40 minutes



Web = HTML + CSS + JS

Язык	Роль	Аналогия
HTML	Структура	Стены и комнаты дома
CSS	Внешний вид	Краска, мебель
JavaScript	Поведение	Свет, двери, лифт

Сегодня — только HTML

 HTML расшифровывается как **HyperText Markup Language** — язык разметки гипертекста. Он существует с **1991 года**.



Как выглядит HTML?

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Page</title>
  </head>
  <body>
    <h1>Hello, Cardiff! 🙌</h1>
    <p>This is my first webpage.</p>
  </body>
</html>
```

💡 **Попробуйте сейчас:** откройте браузер → нажмите **Ctrl+U** → вы увидите HTML любого сайта!



Anatomy of an HTML Tag

открывающий тег

↓
<p class="intro">

↑
атрибут

закрывающий тег

↓
Hello! </p>

↑
содержимое

Два вида тегов:

<!-- Парные – имеют открывающий и закрывающий тег -->

<h1>Title</h1>

<p>Paragraph</p>

Bold text

<!-- Одиночные (void elements) – нет содержимого -->

<hr />

<input type="text" />



HTML Page Structure

```
<!DOCTYPE html>
<!-- версия HTML (всегда HTML5) -->
<html lang="en">
  <!-- корневой элемент, язык -->

  <head>
    <!-- невидимая часть -->
    <meta charset="UTF-8" />
    <!-- кодировка (всегда UTF-8) -->
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Cardiff University</title>
    <!-- название вкладки -->
    <link rel="stylesheet" href="style.css" />
  </head>

  <body>
    <!-- всё что видит пользователь -->
    <h1>Welcome!</h1>
    <p>Content goes here.</p>
  </body>
</html>
```



Text — Headings & Paragraphs

```
<h1>Main Title (only one per page!)</h1>
<h2>Section Title</h2>
<h3>Subsection</h3>
<h4>Sub-subsection</h4>
<h5>Rarely used</h5>
<h6>Almost never used</h6>

<p>
  This is a paragraph. Browsers add space above and below automatically.
  Multiple spaces are collapsed to one.
</p>

<p>This is a second paragraph. Each &lt;p> starts on a new line.</p>

<!-- Line break (use sparingly!) -->
<p>Line one.<br />Line two.</p>

<!-- Horizontal rule -->
<hr />
```



Text Formatting

```
<!-- Semantic (recommended) -->
<strong>Important text</strong>
<!-- bold + meaning -->
<em>Emphasised text</em>
<!-- italic + meaning -->
<mark>Highlighted text</mark>
<!-- yellow highlight -->
<del>Deleted text</del>
<!-- strikethrough -->
<ins>Inserted text</ins>
<!-- underline -->
<small>Small print</small>
<sub>H<sub>2</sub>O</sub>
<!-- subscript -->
<sup>E = mc<sup>2</sup></sup>
<!-- superscript -->
<code>console.log("Hello")</code>
<!-- inline code -->

<!-- Non-semantic (avoid) -->
<b>Just bold</b>
<i>Just italic</i>
```



Lists

```
<!-- Unordered list (bullets) – order doesn't matter -->
```

```
<ul>  
  <li>HTML</li>  
  <li>CSS</li>  
  <li>JavaScript</li>  
</ul>
```

```
<!-- Ordered list (numbers) – order matters -->
```

```
<ol>  
  <li>Open VS Code</li>  
  <li>Create index.html</li>  
  <li>Write your HTML</li>  
  <li>Open in browser</li>  
</ol>
```

```
<!-- Nested list -->
```

```
<ul>  
  <li>  
    Frontend  
    <ul>  
      <li>HTML</li>  
      <li>CSS</li>  
    </ul>  
  </li>  
  <li>Backend</li>  
</ul>
```


Links

```
<!-- External link -->
<a href="https://www.cardiff.ac.uk">Cardiff University</a>

<!-- Open in new tab -->
<a href="https://google.com" target="_blank" rel="noopener"> Google </a>

<!-- Internal link (same website) -->
<a href="about.html">About Us</a>
<a href="/contact">Contact</a>
<!-- from root -->
<a href="../index.html">Home</a>
<!-- one folder up -->

<!-- Link to section on same page -->
<a href="#contact-section">Jump to Contact</a>
<h2 id="contact-section">Contact</h2>

<!-- Email & phone -->
<a href="mailto:hello@cardiff.ac.uk">Email Us</a>
<a href="tel:+442920874000">Call Us</a>
```



Images

```
<!-- Basic image -->


<!-- Image from the internet -->


<!-- With size -->


<!-- Image as a link -->
<a href="https://cardiff.ac.uk">
  
</a>
```

⚠ Always write **alt** text!

- Screen readers read it aloud for blind users
- Shows when image fails to load



Tables

```
<table>
  <thead>
    <!-- заголовок таблицы -->
    <tr>
      <!-- table row -->
      <th>Name</th>
      <!-- table header cell -->
      <th>Course</th>
      <th>Year</th>
    </tr>
  </thead>
  <tbody>
    <!-- тело таблицы -->
    <tr>
      <td>Alice</td>
      <!-- table data cell -->
      <td>Computer Science</td>
      <td>1</td>
    </tr>
    <tr>
      <td>Bob</td>
      <td>Web Development</td>
      <td>2</td>
    </tr>
  </tbody>
</table>
```



Forms

```
<form action="/submit" method="POST">
  <label for="name">Your Name:</label>
  <input type="text" id="name" name="name" placeholder="John Smith" required />

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required />

  <label for="course">Course:</label>
  <select id="course" name="course">
    <option value="">-- Choose --</option>
    <option value="cs">Computer Science</option>
    <option value="web">Web Development</option>
  </select>

  <label for="message">Message:</label>
  <textarea id="message" name="message" rows="4"></textarea>

  <button type="submit">Send</button>
</form>
```



Semantic HTML

```
<!-- ❌ Before HTML5 – everything was a <div> -->
<div id="header">...</div>
<div id="nav">...</div>
<div id="main">...</div>
<div id="footer">...</div>
```

```
<!-- ✅ Modern HTML5 – meaningful tags -->
```

```
<header>
  <nav>
    <a href="/">Home</a>
    <a href="/about">About</a>
  </nav>
</header>

<main>
  <article>
    <h1>Article Title</h1>
    <p>Content...</p>
  </article>
  <aside>Related links...</aside>
</main>

<footer>© 2025 Cardiff University</footer>
```



div and span

```
<!-- div – block container (takes full width) -->
<div class="card">
  <h2>Cardiff Castle</h2>
  <p>A historic castle in the city centre.</p>
</div>

<!-- span – inline container (stays in line) -->
<p>The price is <span class="price">£29.99</span> per person.</p>
```

Block vs Inline

Block	Inline
div, p, h1 – h6	span, a, strong
ul, ol, table	em, img, code
Starts on new line	Stays in text flow
Takes full width	Takes only content width

⚙ Attributes & Comments

```
<!-- Global attributes – work on ANY tag -->
<p id="intro">Unique ID – only one per page</p>
<p class="highlight">CSS class – reusable</p>
<p class="highlight large bold">Multiple classes</p>
<p hidden>This is invisible</p>
<p title="Tooltip text">Hover over me</p>
<p lang="cy">Cymraeg</p>

<!-- data-* attributes – custom data for JavaScript -->
<button data-product-id="42" data-price="9.99">Add to Cart</button>

<!-- Comments – invisible to users, only in source code -->
<!-- This is a comment -->

<!--
  Multi-line comment
  Great for TODO notes
-->
```



Full Page Example

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Cardiff Blog</title>
  </head>
  <body>
    <header>
      <nav>
        <a href="/">Home</a> | <a href="/about">About</a> |
        <a href="/contact">Contact</a>
      </nav>
    </header>
    <main>
      <article>
        <h1>Welcome to Cardiff 🏰</h1>
        
        <p>Cardiff is the <strong>capital city</strong> of Wales.</p>
        <a href="https://visitcardiff.com" target="_blank"> Visit Cardiff → </a>
      </article>
    </main>
    <footer>
      <p>© 2025 Cardiff University</p>
    </footer>
  </body>
</html>
```


✓ Key Takeaways

- 🏷️ **Tags** wrap content: `<tag>content</tag>`
- 📄 **Structure:** `<!DOCTYPE>` → `<html>` → `<head>` + `<body>`
- 📝 **Text:** `<h1>` — `<h6>`, `<p>`, `<strong>`, `<em>`
- 🔗 **Links:** `<a href="...">` — always add `rel="noopener"` for `_blank`
- 🖼️ **Images:** `` — alt is required
- 📋 **Lists:** `<ul>` unordered, `<ol>` ordered, `<li>` items
- 🧠 **Semantic:** use `<header>`, `<main>`, `<article>`, `<footer>`
- 📦 **Block vs Inline:** `<div>` vs `<span>`

Next lesson: **CSS** — making it look good 🎨



Homework

Build your personal profile page:

```
profile/  
└─ index.html
```

1. ✓ Correct `<!DOCTYPE html>` structure
2. ✓ Your name in `<h1>`, course in `<h2>`
3. ✓ A short paragraph `<p>` about yourself
4. ✓ A list `<ul>` of your hobbies
5. ✓ A link `<a>` to your favourite website
6. ✓ At least one image `<img>` with `alt` text
7. ✓ Use semantic tags: `<header>`, `<main>`, `<footer>`

Submit via CodePen or as a `.html` file by next week.



Questions?

Resources

- **MDN Web Docs** — developer.mozilla.org
- **HTML Tutorial** — w3schools.com/html
- **Try it live** — codepen.io
- **Validate your HTML** — validator.w3.org

Next week: CSS — Styling Your Page